

As mentioned in the previous post, the goal is to develop a numerical solver for the unsteady incompressible Navier–Stokes equations in generalized coordinates. As an initial step, the focus is placed on laminar incompressible flow in order to keep the mathematical model as simple as possible and concentrate on the development and verification of the numerical solver.

Assuming the flow to be two-dimensional, laminar, incompressible (constant density), and with constant viscosity significantly simplifies the governing equations. In the laminar regime, the flow is resolved directly and no additional turbulence modelling equations are required, unlike in turbulent simulations where extra transport equations (e.g., for turbulent kinetic energy or dissipation) must be solved. The two-dimensional assumption removes the need for the third momentum equation and by neglecting thermal effects, the energy equation does not need to be solved, which reduces both the number of governing equations and the overall computational complexity.

However, incompressibility introduces a fundamental challenge: since the density is constant, the thermodynamic relation that links pressure and density in compressible flows can no longer be used to determine the pressure field.

One among many solutions is to derive an equation for pressure that enforces the incompressibility constraint. Combining the continuity equation with the momentum equations leads to a pressure Poisson equation. A common numerical strategy that uses this equation to enforce the divergence-free condition is the fractional-step method, in which the velocity and pressure fields are solved in a decoupled manner. First, the momentum equations are solved using a provisional pressure field to obtain an intermediate velocity field. This velocity field generally does not satisfy the incompressibility constraint. The pressure Poisson equation is then solved, and the velocity field is corrected using the resulting pressure gradient so that the updated velocity satisfies the divergence-free condition.

At low Reynolds numbers, where viscous effects dominate the flow, explicit time discretization of the momentum equations becomes severely restricted by stability conditions. In particular, the diffusion stability limit often requires very small time steps, making explicit schemes computationally inefficient. One of the solutions to this problem is to use a semi implicit scheme where convection terms are treated explicitly while diffusion terms are treated implicitly.

To test these ideas, I began by writing a Python code implementing a semi-implicit scheme for the incompressible Navier–Stokes equations using the fractional-step (projection) method. The solver was first applied to the classical lid-driven cavity problem, a widely used benchmark in CFD. Despite its simple geometry, the problem captures the essential physics of incompressible viscous flows, while its geometric simplicity allows the focus to remain on the numerical method itself without the additional complications of metric transformations or complex geometry handling.